Woongbi Cho, Ph.D.
Department of Mechanical Engineering, Stanford University, 440 Escondido Mall, Stanford, CA 94305, United States
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Career Objective

I aim to pursue an academic career focused on developing next-generation polymer composites by integrating multifunctional materials with scalable manufacturing, with an emphasis on processing–structure–property–performance (PSPP) relationships for soft robotics, reconfigurable devices, optoelectronics, and nanogenerators.

Professional Experience

05/2026–Present **Stanford University** Stanford, CA, United States
Postdoctoral Scholar in Mechanical Engineering
- Advisor: Prof. Ruike Renee Zhao, Soft Intelligent Materials Laboratory (SIM Lab) [web]
- Designing electromagnetic (EM)-driven soft robotics, polymer assembly mechanisms, and active metamaterials.

03/2025–04/2026 **Hanyang University** Seoul, Republic of Korea
Postdoctoral Research Fellow in Organic and Nano Engineering
- Advisor: Prof. Jeong Jae (JJ) Wie, Advanced Materials Research Laboratory (AMRL) [web]
- Designing liquid crystalline (LC) polymers composites for soft actuators/robotics and shape-reconfigurable devices, as well as sulfur-rich polymer composites for sustainable and high-performance infrared (IR) optics and triboelectric nanogenerators (TENGs).

Education

09/2022–02/2025 **Hanyang University** Seoul, Republic of Korea
Ph.D. in Organic and Nano Engineering (GPA: 4.22/4.5)
- Advisor: Prof. Jeong Jae (JJ) Wie
- Dissertation title: *"Liquid Crystalline Polymer Composites-Based 3D Shape-Reconfigurable Platform of Locomotive Electronics"*

03/2019–08/2022 **Inha University** Incheon, Republic of Korea
Ph.D. studies in Program in Environmental and Polymer Engineering (GPA: 4.44/4.5)
- Advisor: Prof. Jeong Jae (JJ) Wie
- Transferred to Hanyang University in *Sep 2022* following Prof. JJ Wie's relocation

03/2013–02/2019 **Inha University** Incheon, Republic of Korea
B.S. in Polymer Science and Engineering (GPA: 4.03/4.5)
- Graduated ***Magna cum laude***
- Mandatory Military Service, *03/2015–12/2016*

Awards and Honors

1. Wiley Top Viewed Article, ***Advanced Materials*** (05/2026)
2. Prof. Choi Sam Kwon Ph.D. Dissertation Award, ***Polymer Society of Korea (PSK)*** (04/2025)
3. Outstanding Research Achievement Award (Conference Presentation Division), ***Hanyang University*** (03/2025)
4. Ph.D. Dissertation Award, ***Hanyang University*** (02/2025)
5. Ph.D. Dissertation Award, ***The Research Institute of Industrial Science, Hanyang University*** (02/2025)
6. Excellent Student Oral Presentation Award, ***The 7th International Conference on Active Materials and Soft Mechatronics*** *(2024 AMSM)* (10/2024)
7. Excellent Oral (English) Presentation Award, ***2024 Polymer Society of Korea (PSK)*** *Spring Meeting* (04/2024)
8. Outstanding Research Achievement Award (Publication Division), ***Hanyang University*** (02/2024)
9. Brain Korea (BK) 21 Capstone Design-Excellent Award, ***Hanyang University*** (02/2024)
10. Outstanding Discussant Award, ***The 24th Korea Liquid Crystal Conference*** (01/2024)
11. Honorable Mention (Poster presentation), ***2023 Polymer Society of Korea (PSK) Fall Meeting*** (10/2023)

12. Best Poster Award Nominee, ***2023 Materials Research Society (MRS) Spring Meeting & Exhibit*** (04/2023)
13. Excellent Oral Presentation Award, ***The 23rd Korea Liquid Crystal Conference*** (02/2023)
14. Best Poster Award, ***The 6th International Conference on Advanced Electromaterials (ICAE)*** (11/2021)
15. Best Oral Presentation Award, ***2021 Polymer Society of Korea (PSK) Spring Meeting*** (04/2021)

Professional Service

1. Ad hoc reviewer for Nature Communications, Materials Today, Materials Horizons, Sensors and Actuators B: Chemical, Laser & Photonics Reviews, Macromolecular Bioscience, Micromachines, Applied Sciences, and Nanomaterials.

First Author Publications

1. **W. Cho†**, M. J. Bak†, S. Kim, M. J. Hahm, and J. J. Wie*, "Deborah-Number-Governed Surface Roughness Engineering of Sulfur-Rich Polymer Film for High-Performance Triboelectric Nanogenerators", ***To be submitted***
2. S. B. Park†, **W. Cho†**, M. Lee, S. Kim, G. Y. Bae, N.-H. You, and J. J. Wie*, "Thermally Stable Bulk Longwave Infrared Windows from Sulfur-Rich Polymers for Aerial Thermal Imaging via Kinetics-Controlled Vacuum-Assisted Thermal Treatment", ***To be submitted*** (**Invited article**)
3. M. J. Hahm†, **W. Cho†**, J. Jeon, H.-R. Kim, T. Zhang, and J. J. Wie*, "Spatially Patterned Stiffness Variation in a Light-Triggered Jumper for Symmetry Breaking and High Snap-Through Efficiency", ***Sci. Adv.***, 11 (35), eadx8301 (2025) (**Featured in TechXplore**), **IF: 11.7**
4. **W. Cho†**, S. Kim†, H. Lee, N. Han, H. Kim, M. Lee*, T. H. Han*, and J. J. Wie*, "High-Performance Yet Sustainable Triboelectric Nanogenerator Based on Sulfur-Rich Polymer Composites with MXene-Segregated Structure", ***Adv. Mater.***, 36 (44), 2404163 (2024) (**Back Cover Article**), **IF: 29.4**
5. **W. Cho**, S. Kim, and J. J. Wie*, "Value-Addition of the Waste from Petroleum Refining Process: Sulfur-Rich Polymer for Sustainable and High-Performance Optical and Energy Applications", ***Acc. Mater. Res.***, 5 (5), 625-639 (2024) (**Invited Review, Supplementary Cover Article**), **IF: 14.7**
6. **W. Cho†**, D. J. Kang†, M. J. Hahm, J. Jeon, D.-G. Kim, Y. S. Kim, T. H. Han*, and J. J. Wie*, "Multi-Functional Locomotion of Collectively Assembled Shape-Reconfigurable Electronics", ***Nano Energy***, 118, 108953 (2023) (**Featured in Phys.org**), **IF: 19.069**
7. S. Jhang†, **W. Cho†**, S. K. Lee, A. R. Yu, J. G. Lee, K. Jung, Y. Pu, C. G. Yoo, and J. J. Wie*, "Functional Group Effect of Chemically Modified Microcrystalline Methyl Cellulose on Thermoplastic Polyurethane Composites", ***Cellulose***, 30, 6917-6931 (2023), **IF: 6.123**
8. J. H. Hwang†, S. H. Kim†, **W. Cho†**, W. Lee, S. Park, Y. S. Kim, J.-C. Lee, K. J. Lee*, J. J. Wie*, D.-G. Kim*, "A Microphase Separation Strategy for the Infrared Transparency-Thermomechanical Property Conundrum in Sulfur-Rich Copolymers", ***Adv. Optical Mater.***, 11 (5), 2202432 (2023) (**Inside Front Cover Article**), **IF: 10.050**
9. **W. Cho†**, J. Hwang†, S. Y. Lee†, J. Park, N. Han, C. H. Lee, S.-W. Kang, A. Urbas, J. O. Kim, Z. Ku*, J. J. Wie*, "Highly Sensitive and Cost-Effective Polymeric-Sulfur-Based Mid-Wavelength Infrared Linear Polarizers with Tailored Fabry–Pérot Resonance", ***Adv. Mater.***, 35 (7), 2209377 (2023) (**Featured in Phys.org**), **IF: 32.086**
10. N. Han†, **W. Cho†**, J. H. Hwang, S. Won, D.-G. Kim, and J. J. Wie*, "Enhancement of Thermomechanical Properties of Sulfur-Rich Polymers by Post-Thermal Treatment", ***Polym. Chem.***, 14 (8), 943 (2023) (**Invited article, Themed Collection: Chalcogen-rich polymers**), **IF: 5.364**
11. **W. Cho†**, J. Jeon†, W. Eom, J. G. Lee, D.-G. Kim, Y. S. Kim, T. H. Han*, and J. J. Wie*, "Photo-Triggered Shape Reconfiguration in Stretchable Reduced Graphene Oxide-Patterned Azobenzene-Functionalized Liquid Crystalline Polymer Networks", ***Adv. Funct. Mater.***, 31 (31), 2102106 (2021) (**Featured in Phys.org**), **IF: 16.836**
12. **W. Cho**, D. Y. Hahm, J. H. Yim, J. H. Lee, Y. J. Lee, D.-G. Kim, Y. S. Kim, and J. J. Wie*, "Programmable Building blocks via Internal Stress Engineering for 3D Collective Assembly", ***Adv. Mater. Technol.***, 5 (12), 2000758 (2020) (**Front Cover Article**), **IF: 5.395**

Co-Author Publications

13. M. J. Hahm, **W. Cho**, J. Jeon, K. Yang, and J. J. Wie*, "Physically Intelligent Liquid Crystalline Polymers for Soft Robots and Shape-Reconfigurable Devices", ***ACS Appl. Opt. Mater.***, 4 (4), 962-974 (2026) (**Invited Spotlight article**)
14. J. H. Hwang†, S. Won†, J. M. Lee, **W. Cho**, S. Park, H. Kim, C.-G. Chae, W. Lee, D.-G. Kim*, J. J. Wie*, and Y. S. Kim*, "Closed-Loop and Sustainable 4D Printing of Multi-Stimuli-Responsive Sulfur-Rich Polymer Composites for Autonomous Task Execution", ***Adv. Mater.***, 37 (44), e07057 (2025) (**Front Cover Article**), **IF: 27.4**
15. Y. Yoon†, H. Moon†, **W. Cho**, D. Lee, S. Jeong*, J. J. Wie*, and C. B. Kim*, "Light-Fueled In-Operando Shape

Reconfiguration, Fixation, and Recovery of Magnetically Actuated Microtextured Covalent Adaptable Networks", ***Adv. Mater.***, 37 (39), 2503161 (2025) (**Front Cover Article**), **IF: 27.4**

16. J. Molineux, S. W. Durfee, R. Hyde, M. Malaker, J. P. Narr, K.-J. Kim, R. Himmelhuber, Y.-J Kim, M.-G. Bog, B. Jeong, **W. Cho**, J. J. Wie, N. B. Krishnan, Y. Y. Yang, R. A. Norwood*, and J. Pyun*, "Disulfide Glass: An Optical Polymer for Commodity Plastics, Precision Optics, and Photonics", ***Adv. Funct. Mater.***, 35 (22), 2422569 (2025), **IF: 18.5**
17. C. Olikagu, S. Khoshsorour, S. D. Dulam, H.-S. Yu, N. A. Graham, K.-J. Kim, B. Jeong, J. L. Hedrick, A. B.-Stoner, K. Cheng, B. L. Batchelor, **W. Cho**, S. B. Park, J. J. Wie, Y.-J. Kim, M.-G. Bog, N. B. Krishnan, Y. Y. Yang, J. T. Njardarson, R. A. Norwood*, and J. Pyun*, "Photopolymer Resins from Sulfenyl Chloride Commodity Chemicals for Plastic Optics, Photopatterning and 3D-printing" ***Adv. Mater.***, 37 (14), 2418149 (2025), **IF: 27.4**
18. M. Zhang, S. Jeong, **W. Cho**, J. Ryu, B. Zhang, P. Crovella, A. J. Ragauskas, J. J. Wie*, and C. G. Yoo*, "Green co-solvent-assisted one-pot synthesis of high-performance flexible lignin polyurethane foam", ***Chem. Eng. J.***, 499, 156142 (2024), **IF: 15.1**
19. H. Moon†, J. G. Lee†, **W. Cho**, J. Jeon, J. E. Park, and J. J. Wie*, "Structure-Property-Actuation Relationships of Shape-Fixable Magnetic Vitrimer Micropillar Arrays", ***Sens. Actuators B Chem.***, 402, 135092 (2024), **IF: 8.4**
20. J. E. Park, H. Je, C. R. Kim, S. Park, Y. Yu, **W. Cho**, S. Won, D. J. Kang, T. H. Han, R. Kwak, S. G. Lee*, S. Kim*, and J. J. Wie*. "Programming Anisotropic Functionality of 3D Microdenticles by Staggered-Overlapped and Multi-Layered Microarchitectures", ***Adv. Mater.***, 36 (7), 2309518 (2024) (**Featured in Phys.org**), **IF: 29.4**
21. H. Moon, J. E. Park, **W. Cho**, J. Jeon, and J. J. Wie*, "Curing Kinetics and Structure-Property Relationship of Moisture-Cured One-Component Polyurethane Adhesives", ***Eur. Polym. J.***, 201, 112579 (2023), **IF: 5.546**
22. J.-C. Choi†, J. Jeon†, J.-W. Lee, A. Nauman, J. G. Lee, **W. Cho**, C. Lee, Y.-M. Cho, J. J. Wie*, and H.-R. Kim*, "Steerable and Agile Light-Fueled Rolling of Soft Robots by Curvature-Engineered Torsional Torque", ***Adv. Sci.***, 10 (30), 2304715 (2023) (**Front Cover Article**), **IF: 17.521**
23. K. Lee†, J. Jeon†, **W. Cho**, S.-W. Kim, H. Moon, J. J. Wie* and S. Lee*, "Light-triggered Autonomous Shape-reconfigurable and Locomotive Rechargeable Power Sources", ***Mater. Today***, 55, 56-65 (2022) (**Featured in Materials Today.com**), **IF: 31.041**
24. J. Jeon, H. Choi, **W. Cho**, J. Hong, J. Youk, and J. J. Wie*, "Height-Tunable Replica Molding Using Viscous Polymeric Resins", ***ACS Macro Lett.***, 11 (4), 428-433 (2022) (**Supplementary Cover Article**), **IF: 6.903**
25. J. Jeon†, J. C. Choi†, H. Lee, **W. Cho**, K. Lee, J. G. Kim, J. W. Lee, K. I. Joo, M. Cho*, H. R. Kim*, J. J. Wie*, "Continuous and Programmable Photomechanical Jumping of Polymer Monoliths", ***Mater. Today***, 49, 97-106 (2021) (**Inner Cover Article, Featured in Phys.org**), **IF:26.416**
26. J. E. Park, S. Won, **W. Cho**, J. G. Kim, S. Jhang, J. G. Lee, and J. J. Wie*, "Fabrication and Applications of Stimuli-Responsive Micro/Nanopillar Arrays", ***J. Polym. Sci.***, 59 (14), 1491-1517 (2021) (**Invited Review**)

Patents

Granted Patents

1. J. J. Wie, Z. Ku, **W. Cho**, and S. Y. Lee, "Mid-Wavelength Infrared Polarizer Based on Sulfur-Containing Polymer and Manufacturing Method Thereof", Patent No. 10-2940899, Republic of Korea (2026)
2. J. J. Wie, **W. Cho**, and N. Han "Preparation Method of Sulfur-Rich Polymer Using Inverse Vulcanization and Vacuum Assisted Thermal Treatment, and Sulfur-Rich Polymer Preparing Thereby", Patent No. 10-2957615, Republic of Korea (2026)

Pending Patent Applications

1. J. J. Wie, T. Han, **W. Cho**, and S. Kim, "Sulfur-Containing Polymer Composites, Preparation Method and Triboelectric Generator Using The Same", Patent application No. 10-2024-0073750, Republic of Korea (2024)
2. J. J. Wie, **W. Cho**, and S. Park, "Method for vacuum-assisted thermal treatment of sulfur-rich polymers with particle size control and infrared optical article manufactured thereby", Patent application No. 10-2026-0049911, Republic of Korea (2026)
3. C. B. Kim, J. J. Wie, S. Jeong, Y. Yoon, H. Moon, **W. Cho**, and D. W. Lee "A magnetically actuated microtextured polymer composite structure, method for manufacturing the same, and method for shape reconfiguration thereof", Patent application No. 10-2026-0064515, Republic of Korea (2026)
4. C. B. Kim, J. J. Wie, S. Jeong, Y. Yoon, H. Moon, **W. Cho**, and D. W. Lee "Polymer Composite Structure Having Magnetically Driven Microstructure, Method of Manufacturing the Same, and Method of Transforming the Shape Thereof,", U.S. Patent application No. 19.687,112 filed May 26, 2026

Invited Talks and Seminars

1. Korea Research Institute of Chemical Technology (KRICT), Advanced Functional Polymers Research Center, Republic of Korea (03/2026)
   - Title: "Processing–Structure–Properties Relationships in Polymer Composites: From Commercial Thermoplastics to Multi-functional Polymer Systems"
2. Korea Liquid Crystal Conference, Young Investigator Oral Presentation, Republic of Korea (11/2025)
   - Title: "Liquid Crystalline Polymer/2D material Bilayers for Electrically Conductive, Shape-Reconfigurable Systems"

Selected Conference Presentations

1. **W. Cho**, S. Kim, H. Lee, N. Han, H. Kim, M. Lee, T. H. Han, and J. J. Wie*, "Segregated Structure Engineered Sulfur-Rich Polymer/MXene Composites for High-Performance Triboelectric Nanogenerator", ***American Chemical Society (ACS) Fall 2025, Washington D.C.*** (08/2025)
2. **W. Cho**, D. J. Kang, M. J. Hahm, J. Jeon, D.-G. Kim, Y. S. Kim, T. H. Han, and J. J. Wie*, "Collectively Architectured Liquid Crystalline Elastomer/MXene Composites for Multi-functional Shape-Reconfigurable Electronics", **The 7th International Conference on Active Materials and Soft Mechatronics (2024 AMSM)** (10/2024)
   - Excellent Student Oral Presentation Award
3. **W. Cho**, S. Kim, H. Lee, N. Han, H. Kim, M. Lee, T. H. Han, and J. J. Wie*, "High-Performance Triboelectric Nanogenerator Based on Sulfur-Rich Polymer/MXene Composites for Value-Addition of Waste Sulfur", ***American Chemical Society (ACS) Fall 2024, Denver*** (08/2024)
4. **W. Cho**, S. Kim, H. Lee, N. Han, H. Kim, M. Lee, T. H. Han, and J. J. Wie*, "Sustainable and High-Performance Triboelectric Nanogenerator Based on Sulfur-Rich Polymer/MXene Composites" ***2024 Materials Research Society (MRS) Spring Meeting & Exhibit, Seattle*** (04/2024)
5. **W. Cho**, S. Kim, H. Lee, N. Han, H. Kim, M. Lee, T. H. Han, and J. J. Wie*, "Reusable yet High-Performance Triboelectric Nanogenerators by Sulfur-Rich Polymer/MXene Composites" ***2024 Polymer Society of Korea (PSK) Spring Meeting*** (04/2024)
   - Excellent Student Oral (English) Presentation Award
6. **W. Cho**, J. Jeon, M. J. Hahm, T. H. Han, and J. J. Wie*, "3D Shape-Reconfigurable and Locomotive Electronics Based on Physical Intelligence-Encoded Polymer Composites" ***2023 Materials Research Society (MRS) Fall Meeting & Exhibit, Boston*** (11/2023)
7. **W. Cho**, J. Hwang, S. Y. Lee, Z. Ku, and J. J. Wie*, "Highly Sensitive Mid-Wavelength Infrared Linear Polarizers Based on Waste from Petroleum Refining Process", ***2023 Materials Research Society (MRS) Fall Meeting & Exhibit, Boston*** (11/2023)
8. **W. Cho**, S. Jhang, S. K. Lee, and J. J. Wie*, "Functional Group Effects of Micro Crystalline Methyl Cellulose on Eco-Polymer Composites", ***2023 Polymer Society of Korea (PSK) Fall Meeting*** (10/2023)
   - Honorable Mention (Poster presentation)
9. **W. Cho**, H. Lee, S. Kim, M. Lee, T. H. Han, and J. J. Wie*, "2D MXene Embedded Sulfur-Rich Polymer for High-Performance Triboelectric Negative Material", ***2023 Materials Research Society (MRS) Spring Meeting & Exhibit, San Francisco*** (04/2023)
   - Best Poster Award Nominee
10. **W. Cho**, D. J. Kang, M. J. Hahm, T. H. Han, and J. J. Wie*, "Multi-Functional Locomotion of Shape-Reconfigurable Electronics", **The 23rd Korea Liquid Crystal Conference organized by Korean Information Display Society (KiDS)** (02/2023)
    - Excellent Student Oral Presentation Award
11. **W. Cho**, and J. J. Wie*, "Soft-Robotic Actuations of Collectively Assembled Soft-Electronics Based on MXene/Liquid Crystal Elastomer Bilayer", ***2022 Materials Research Society (MRS) Fall Meeting & Exhibit, Boston*** (11/2022)
12. **W. Cho**, J. Hwang, S. Y. Lee, J. Park, N. Han, J. O. Kim, A. Urbas, Z. Ku, and J. J. Wie*, "High optical performance of Polymeric Sulfur Based Mid-Wavelength Infrared Linear Polarizers", ***2022 Materials Research Society (MRS) Spring Meeting & Exhibit, Honolulu*** (05/2022)
13. **W. Cho**, J. Jeon, W. Eom, J. G. Lee, D.-G. Kim, Y. S. Kim, T. H. Han, and J. J. Wie*, "Light-Triggered Shape-Reconfiguration of Programmable and Stretchable Electronics with Enhanced Mechanical and Actuation Performance", ***The 6th International Conference on Advanced Electromaterials (ICAE)*** (11/2021)
    - Best Poster Award (Gold Prize)
14. **W. Cho**, J. Jeon, W. Eom, T. H. Han, and J. J. Wie*, "Multi-Stimuli Responsive Reduced Graphene Oxide Patterned Azo-Liquid Crystalline Polymer Networks for Stretchable Electronics with Tailored Morphologies", ***2021 Virtual Materials Research Society (MRS) Fall Meeting & Exhibit*** (11/2021)

15. **W. Cho**, J. Jeon, W. Eom, J. G. Lee, D.-G. Kim, Y. S. Kim, T. H. Han, and J. J. Wie*, “Stretchable Electronics with Light-Triggered Shape-Reconfiguration: Reduced Graphene Oxide Patterned Azo-Liquid Crystalline Polymer Networks with Enhanced Mechanical and Actuating Performance” ***American Chemical Society (ACS) Fall 2021 National Meeting*** (08/2021)
16. **W. Cho**, J. Jeon, W. Eom, J. G. Lee, D.-G. Kim, Y. S. Kim, T. H. Han, and J. J. Wie*, “Multi-Stimuli Responsive and Electrically Conducting rGO Patterned Azo-LCN for Enhanced Mechanical Stiffness Simultaneously with High Deformability”, **The 48th World Polymer Congress, IUPAC-MACRO2020+ organized by Polymer Society of Korea (PSK)** (05/2021)
17. **W. Cho**, D.-G. Kim, Y. Kim, and J. J. Wie*, “Shape Controllable Building Blocks for 3D Collective Assembly via Residual Stress Engineering”, ***2021 Virtual Materials Research Society (MRS) Spring Meeting & Exhibit*** (04/2021)
18. **W. Cho**, J. Jeon, W. Eom, T. H. Han, and J. J. Wie*, “Shape-reconfigurable rGO patterned azo-LCN with tailored mechanical and electrical properties”, ***2021 Polymer Society of Korea (PSK) Spring Meeting*** (04/2021)
19. **W. Cho**, J. Jeon, W. Eom, T. H. Han, D.-G. Kim, Y. S. Kim, and J. J. Wie*, “Reconfigurable Azo-LCN/rGO nanocomposites with Tailorable Mechanical and Electrical Properties”, ***2020 Virtual Materials Research Society (MRS) Spring/Fall meeting & Exhibit*** (11/2020)

Research Projects

1. Flexible and Wearable Triboelectric Devices Based on Polymeric Sulfur/MXene Composites, U.S. Army International Technology Center Pacific (U.S. Army/USA, 01/2023–12/2024)
2. Magneto-Mechanically Reconfigurable Metasurface, Asian Office of Aerospace Research and Development, U.S. Air Force Office of Scientific Research (AOARD/USA, 09/2022–09/2024)
3. Reversibly Reconfigurable 3D Micro- and Nanophotonic Devices by Magnetically Programmable Polymeric Composites, Asian Office of Aerospace Research and Development, U.S. Air Force Office of Scientific Research (AOARD/USA, 09/2018–09/2021)
4. Development of a 3D Mapping Process for Crosslinking Density to Improve Glass Adhesive Performance, Hyundai Motor Group (Hyundai Motor Group/KOR, 04/2023–06/2024)
5. Development of Curing Behavior and Degree-of-Cure Measurement Methods for Moisture-Curable DGU Glass Adhesives, Hyundai Motor Group (Hyundai Motor Group/KOR, 09/2021–09/2022)
6. Contact Electrification Basic Research Laboratory, National Research Foundation of Korea (NRF/KOR, 07/2020–02/2023)
7. Development of High-Refractive-Index, Infrared-Transparent Sulfur-Containing Optical Polymeric Materials and Application Technologies for the Sensor Market, Ministry of Trade, Industry and Energy and Korea Evaluation Institute of Industrial Technology (MOTIE/KEIT/KOR, 04/2020–12/2024)
8. FDM-Based Multimaterial 4D Printing and Structural Design for Vitrimer Composites, National Research Foundation of Korea (NRF/KOR, 09/2019–12/2022)
9. Liquid Crystal Vitrimer-Based Physical Intelligence-Embedded Toxic Gas Detection Sensor, Korea Research Institute of Chemical Technology (KRICT/KOR, 01/2024–12/2024)
10. Photo/Magneto-Responsive Multifunctional Actuation of Collectively Assembled Liquid Crystalline Polymers, Korea Research Institute of Chemical Technology (KRICT/KOR, 01/2023–12/2023)
11. Programmable Liquid Crystalline Polymer Composites for Multifunctional Soft Actuators, Korea Research Institute of Chemical Technology (KRICT/KOR, 01/2022–12/2022)
12. Stimuli-Responsive 3D Shape-Reconfigurable Electrically Conductive Polymer Composites, Korea Research Institute of Chemical Technology (KRICT/KOR, 01/2021–12/2021)
13. Collective Assembly for 3D Hierarchical Structures through Residual Stress Engineering, Korea Research Institute of Chemical Technology (KRICT/KOR, 01/2020–12/2020)
14. Multi-Stimuli-Responsive Liquid Crystalline Polymer Nanocomposites, Korea Research Institute of Chemical Technology (KRICT/KOR, 01/2019–12/2019)